Machine Learning
Project-01 Data Preprocessing
Objective:

The main objective of this project is to design and implement a robust data preprocessing systen addresses common challenges such as missing values, outliers, inconsistent formatting, and nois performing effective data preprocessing, the project aims to enhance the quality, reliability, and usefulness of the data for machine learning.

1. Acquire the dataset

Dataset : Employee.csv

2. import the libraries

In [44]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

3. import the dataset

In [45]:
```
df=pd.read_csv('/kaggle/input/employee2/Employee.csv')
df.shape
```

Out[45]:
```
(148, 6)
```

In [46]:
```
df
```

Out[46]:

	Company	Age	Salary	Place	Country	Gender
0	TCS	20.0	NaN	Chennai	India	0
1	Infosys	30.0	NaN	Mumbai	India	0
2	TCS	35.0	2300.0	Calcutta	India	0
3	Infosys	40.0	3000.0	Delhi	India	0
4	TCS	23.0	4000.0	Mumbai	India	0
...	...	...	...	...	...	...
143	TCS	33.0	9024.0	Calcutta	India	1
144	Infosys	22.0	8787.0	Calcutta	India	1
145	Infosys	44.0	4034.0	Delhi	India	1
146	TCS	33.0	5034.0	Mumbai	India	1
147	Infosys	22.0	8202.0	Cochin	India	0

148 rows × 6 columns

In [ ]:

Data Exploration

In [47]:
```
# delete duplicate rows
```

In [48]:
```
df.duplicated().sum()
```

Out[48]:
```
np.int64(4)
```

In [49]:
```
df.drop_duplicates(inplace=True)
df.shape
# 4 duplicates removed
```

Out[49]:
```
(144, 6)
```

In [50]:
```
df.reset_index(inplace=True)
df.drop('index',axis=1,inplace=True)
df
```

Out[50]:

	Company	Age	Salary	Place	Country	Gender
0	TCS	20.0	NaN	Chennai	India	0
1	Infosys	30.0	NaN	Mumbai	India	0
2	TCS	35.0	2300.0	Calcutta	India	0
3	Infosys	40.0	3000.0	Delhi	India	0
4	TCS	23.0	4000.0	Mumbai	India	0
...	...	...	...	...	...	...
139	Infosys Pvt Lmt	22.0	8202.0	Mumbai	India	0
140	TCS	33.0	9024.0	Calcutta	India	1
141	Infosys	44.0	4034.0	Delhi	India	1
142	TCS	33.0	5034.0	Mumbai	India	1
143	Infosys	22.0	8202.0	Cochin	India	0

144 rows × 6 columns

In [51]:
```
df.head(5)
```

Out[51]:

	Company	Age	Salary	Place	Country	Gender
0	TCS	20.0	NaN	Chennai	India	0
1	Infosys	30.0	NaN	Mumbai	India	0
2	TCS	35.0	2300.0	Calcutta	India	0
3	Infosys	40.0	3000.0	Delhi	India	0
4	TCS	23.0	4000.0	Mumbai	India	0

In [52]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 144 entries, 0 to 143
Data columns (total 6 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   Company  136 non-null    object
 1   Age      127 non-null    float64
 2   Salary   121 non-null    float64
 3   Place    130 non-null    object
 4   Country  144 non-null    object
 5   Gender   144 non-null    int64
dtypes: float64(2), int64(1), object(3)
memory usage: 6.9+ KB
```

In [53]:
```
#Check if there is any null values
df.isna().sum()
```

Out[53]:
```
Company     8
Age        17
Salary     23
Place      14
Country     0
Gender      0
dtype: int64
```

In [54]:
```
df.columns
```

Out[54]:
```
Index(['Company', 'Age', 'Salary', 'Place', 'Country', 'Gender'], dtype='object')
```

In [55]:
```
#Unique value in each columns
```

In [56]:
```
unique=df.nunique()
unique
```

Out[56]:
```
Company     6
Age        29
Salary     40
Place      11
Country     1
Gender      2
dtype: int64
```

In [57]:
```
df['Company'].unique()
```

Out[57]:
```
array(['TCS', 'Infosys', 'CTS', nan, 'Tata Consultancy Services',
       'Congnizant', 'Infosys Pvt Lmt'], dtype=object)
```

In [58]:
```
# observations: here 'Infosys' & 'Infosys Pvt Lmt','TCS' & 'Tata Consultancy Ser
# are same.
# Treat them appropriately.
```

In [59]:
```
df['Company'].replace('Infosys Pvt Lmt','Infosys',inplace=True)
df['Company'].replace('Tata Consultancy Services','TCS',inplace=True)
df['Company'].replace('CTS','Congnizant',inplace=True)
df['Company'].replace('Congnizant','Cognizant',inplace=True)
df['Company'].unique()
```

```
/tmp/ipykernel_55/4075095943.py:1: FutureWarning: A value is trying to be set on a co
a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because t
intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({
value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the ope
inplace on the original object.


  df['Company'].replace('Infosys Pvt Lmt','Infosys',inplace=True)
/tmp/ipykernel_55/4075095943.py:2: FutureWarning: A value is trying to be set on a co
a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because t
intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({
value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the ope
inplace on the original object.


  df['Company'].replace('Tata Consultancy Services','TCS',inplace=True)
/tmp/ipykernel_55/4075095943.py:3: FutureWarning: A value is trying to be set on a co
a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because t
intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({
value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the ope
inplace on the original object.


  df['Company'].replace('CTS','Congnizant',inplace=True)
/tmp/ipykernel_55/4075095943.py:4: FutureWarning: A value is trying to be set on a co
a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because t
intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({
value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the ope
inplace on the original object.


  df['Company'].replace('Congnizant','Cognizant',inplace=True)
```

Out[59]:
```
array(['TCS', 'Infosys', 'Cognizant', nan], dtype=object)
```

In [60]:
```
df['Age'].unique()
```

Out[60]:
```
array([20., 30., 35., 40., 23., nan, 34., 45., 18., 22., 32., 37., 50.,
       21., 46., 36., 26., 41., 24., 25., 43., 19., 38., 51., 31., 44.,
       33., 17.,  0., 54.])
```

In [61]:
```
df['Salary'].unique()
```

Out[61]:
```
array([  nan, 2300., 3000., 4000., 5000., 6000., 7000., 8000., 9000.,
       1089., 1234., 3030., 3045., 3184., 4824., 5835., 7084., 8943.,
       8345., 9284., 9876., 2034., 7654., 2934., 4034., 5034., 8202.,
       9024., 4345., 6544., 6543., 3234., 4324., 5435., 5555., 8787.,
       3454., 5654., 5009., 5098., 3033.])
```

In [62]:
```
df['Place'].unique()
# spelling correction for 'Podicherry'.
```

Out[62]:
```
array(['Chennai', 'Mumbai', 'Calcutta', 'Delhi', 'Podicherry', 'Cochin',
       nan, 'Noida', 'Hyderabad', 'Bhopal', 'Nagpur', 'Pune'],
      dtype=object)
```

In [63]:
```
df['Place'].replace('Podicherry','Pondicherry',inplace=True)
df['Place'].unique()
```

```
/tmp/ipykernel_55/124090606.py:1: FutureWarning: A value is trying to be set on a cop
DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because t
intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({
value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the ope
inplace on the original object.


  df['Place'].replace('Podicherry','Pondicherry',inplace=True)
```

Out[63]:
```
array(['Chennai', 'Mumbai', 'Calcutta', 'Delhi', 'Pondicherry', 'Cochin',
       nan, 'Noida', 'Hyderabad', 'Bhopal', 'Nagpur', 'Pune'],
      dtype=object)
```

In [64]:
```
df['Country'].unique()
```

Out[64]:
```
array(['India'], dtype=object)
```

In [65]:
```
df['Gender'].unique()
```

Out[65]:
```
array([0, 1])
```

In [66]:
```
#Statistical analysis
```

In [67]:
```
df.describe()
```

Out[67]:

	Age	Salary	Gender
count	127.000000	121.000000	144.000000
mean	30.527559	5283.471074	0.222222
std	11.114717	2585.373600	0.417191
min	0.000000	1089.000000	0.000000
25%	22.000000	3030.000000	0.000000
50%	33.000000	5000.000000	0.000000
75%	37.500000	8000.000000	0.000000
max	54.000000	9876.000000	1.000000

In [68]:
```
df.head(5)
```

Out[68]:

	Company	Age	Salary	Place	Country	Gender
0	TCS	20.0	NaN	Chennai	India	0
1	Infosys	30.0	NaN	Mumbai	India	0
2	TCS	35.0	2300.0	Calcutta	India	0
3	Infosys	40.0	3000.0	Delhi	India	0
4	TCS	23.0	4000.0	Mumbai	India	0

Data Cleaning

In [69]:
```
#Outlier detection
#using IQR
#Chances of getting outliers are only in the 'Age' and 'Salary' columns.
```

In [70]:
```
df1=pd.read_csv('/kaggle/input/employee2/Employee.csv')
```

In [71]:
```
df1.describe()
```

Out[71]:

	Age	Salary	Gender
count	130.000000	124.000000	148.000000
mean	30.484615	5312.467742	0.222973
std	11.096640	2573.764683	0.417654
min	0.000000	1089.000000	0.000000
25%	22.000000	3030.000000	0.000000
50%	32.500000	5000.000000	0.000000
75%	37.750000	8000.000000	0.000000
max	54.000000	9876.000000	1.000000

In [72]:
```
#Outlier detection in 'Age' column=IQR
```

In [73]:
```
q1=df['Age'].quantile(0.25)
q1
```

Out[73]:
```
np.float64(22.0)
```

In [74]:
```
q3=df['Age'].quantile(0.75)
q3
```

Out[74]:
```
np.float64(37.5)
```

In [75]:
```
iqr=q3-q1
iqr
```

Out[75]:
```
np.float64(15.5)
```

In [76]:
```
lower=q1-1.5*iqr
lower
```

Out[76]:
```
np.float64(-1.25)
```

In [77]:
```
upper=q3+1.5*iqr
upper
```

Out[77]:
```
np.float64(60.75)
```

In [78]:
```
#outlier rows
df[(df['Age']<lower) & (df['Age']>upper)]
#no outliers detected using IQR. But Age=0 is an outlier.
```

Out[78]:

Company	Age	Salary	Place	Country	Gender

In [79]:
```
#Outlier detection in 'Salary' column=IQR
```

In [80]:
```
q1=df['Salary'].quantile(0.25)
q1
```

Out[80]:
```
np.float64(3030.0)
```

In [81]:
```
q3=df['Salary'].quantile(0.75)
q3
```

Out[81]:
```
np.float64(8000.0)
```

In [82]:
```
iqr=q3-q1
iqr
```

Out[82]:
```
np.float64(4970.0)
```

In [83]:
```
lower=q1-1.5*iqr
lower
```

Out[83]:
```
np.float64(-4425.0)
```

In [84]:
```
upper=q3+1.5*iqr
upper
```

Out[84]:
```
np.float64(15455.0)
```

In [85]:
```
#outlier rows
df[(df['Salary']<lower) & (df['Salary']>upper)]
#no outliers detected using IQR.
```

Out[85]:

Company	Age	Salary	Place	Country	Gender

In [ ]:
```

```

In [86]:
```
#Treat missing values
```

In [87]:
```
df.isna().sum()
```

Out[87]:
```
Company     8
Age        17
Salary     23
Place      14
Country     0
Gender      0
dtype: int64
```

In [88]:
```
# For the 'Company' column we have rows with 'nan' values. It should be replaced
```

In [89]:
```
df['Company'].mode()[0]
```

Out[89]:
```
'TCS'
```

In [90]:
```
df['Company'].fillna(df['Company'].mode()[0], inplace=True)
```

```
/tmp/ipykernel_55/5373860.py:1: FutureWarning: A value is trying to be set on a copy
DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because t
intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({
value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the ope
inplace on the original object.


  df['Company'].fillna(df['Company'].mode()[0], inplace=True)
```

In [91]:
```
df.isna().sum()
```

Out[91]:
```
Company     0
Age        17
Salary     23
Place      14
Country     0
Gender      0
dtype: int64
```

In [92]:
```
#treat 'Age' column
# replace the value'0' in Age as 'NaN'
# For the 'Age' column, fillna by mean value
```

In [93]:
```
df[df['Age']==0]
```

Out[93]:

	Company	Age	Salary	Place	Country	Gender
87	Infosys	0.0	3030.0	Calcutta	India	0
91	TCS	0.0	3045.0	Delhi	India	0
100	Cognizant	0.0	2034.0	Pondicherry	India	0
106	TCS	0.0	9024.0	Chennai	India	1
110	Infosys	0.0	3234.0	Mumbai	India	0
120	Cognizant	0.0	1234.0	Calcutta	India	0

In [94]:
```
df['Age'].replace(0,np.nan,inplace=True)
```

```
/tmp/ipykernel_55/771178559.py:1: FutureWarning: A value is trying to be set on a cop
DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because t
intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({
value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the ope
inplace on the original object.


  df['Age'].replace(0,np.nan,inplace=True)
```

In [95]:
```
df.isna().sum()
```

Out[95]:
```
Company     0
Age        23
Salary     23
Place      14
Country     0
Gender      0
dtype: int64
```

In [96]:
```
df.describe()
#now the minimum value for 'Age' is 17 and max is 54 (no outliers). so replace t
```

Out[96]:

	Age	Salary	Gender
count	121.000000	121.000000	144.000000
mean	32.041322	5283.471074	0.222222
std	8.989250	2585.373600	0.417191
min	17.000000	1089.000000	0.000000
25%	23.000000	3030.000000	0.000000
50%	33.000000	5000.000000	0.000000
75%	40.000000	8000.000000	0.000000
max	54.000000	9876.000000	1.000000

In [97]:
```
rounded_mean=round(df['Age'].mean(),0)
rounded_mean
```

Out[97]:
```
np.float64(32.0)
```

In [98]:
```
df['Age'].fillna(rounded_mean,inplace=True)
```

```
/tmp/ipykernel_55/3436453102.py:1: FutureWarning: A value is trying to be set on a co
a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because t
intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({
value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the ope
inplace on the original object.


  df['Age'].fillna(rounded_mean,inplace=True)
```

In [99]:
```
df.isna().sum()
```

Out[99]:
```
Company     0
Age         0
Salary     23
Place      14
Country     0
Gender      0
dtype: int64
```

In [100…
```
# Treat 'Salary' column
# replace the nulls by mean value
```

In [101…
```
rounded_salary=round(df['Salary'].mean(),0)
rounded_salary
```

Out[101…
```
np.float64(5283.0)
```

In [102…
```
df['Salary'].fillna(rounded_salary,inplace=True)
```

```
/tmp/ipykernel_55/2613942818.py:1: FutureWarning: A value is trying to be set on a co
a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because t
intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({
value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the ope
inplace on the original object.


  df['Salary'].fillna(rounded_salary,inplace=True)
```

In [103…
```
df.isna().sum()
```

Out[103…
```
Company     0
Age         0
Salary      0
Place      14
Country     0
Gender      0
dtype: int64
```

In [104…
```
# Treat 'Place' column
# replace the nulls by mode value
```

In [105…
```
df['Place'].mode()
```

Out[105…
```
0    Mumbai
Name: Place, dtype: object
```

In [106…
```
df['Place'].fillna(df['Place'].mode()[0],inplace=True)
```

```
/tmp/ipykernel_55/1144197863.py:1: FutureWarning: A value is trying to be set on a co
a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because t
intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({
value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the ope
inplace on the original object.


  df['Place'].fillna(df['Place'].mode()[0],inplace=True)
```

In [107…

```
df.isna().sum()
```

Out[107…

```
Company    0
Age        0
Salary     0
Place      0
Country    0
Gender     0
dtype: int64
```

In [108…

```
# Now the dataset df is cleaned
```

In [109…

```
df
```

Out[109…

	Company	Age	Salary	Place	Country	Gender
0	TCS	20.0	5283.0	Chennai	India	0
1	Infosys	30.0	5283.0	Mumbai	India	0
2	TCS	35.0	2300.0	Calcutta	India	0
3	Infosys	40.0	3000.0	Delhi	India	0
4	TCS	23.0	4000.0	Mumbai	India	0
...	...	...	...	...	...	...
139	Infosys	22.0	8202.0	Mumbai	India	0
140	TCS	33.0	9024.0	Calcutta	India	1
141	Infosys	44.0	4034.0	Delhi	India	1
142	TCS	33.0	5034.0	Mumbai	India	1
143	Infosys	22.0	8202.0	Cochin	India	0

144 rows × 6 columns

Data Analysis

In [110…

```
#Filter the data with age >40 and salary<5000
```

In [111…

```
df[(df['Age']>40) & (df['Salary']<5000)]
```

Out[111…

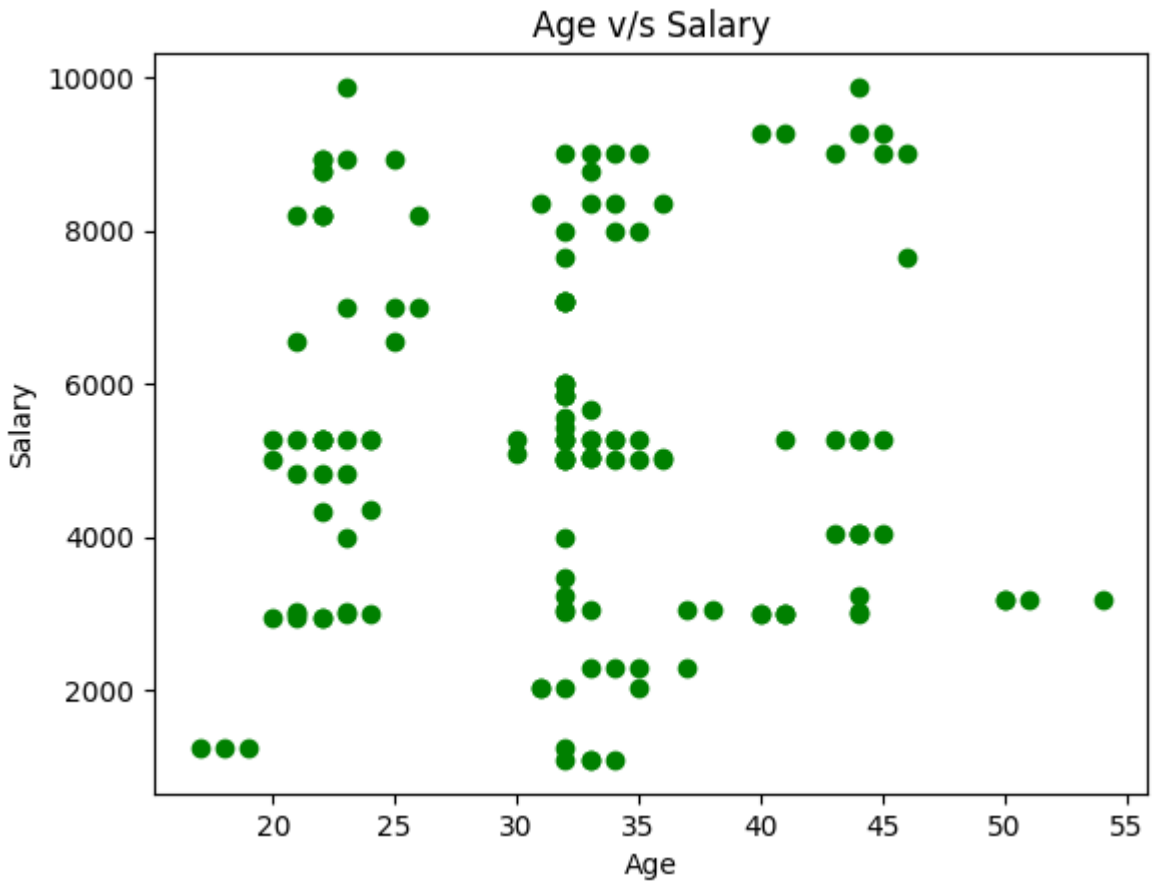
	Company	Age	Salary	Place	Country	Gender
21	Infosys	50.0	3184.0	Delhi	India	0
32	Infosys	45.0	4034.0	Calcutta	India	0
39	Infosys	41.0	3000.0	Mumbai	India	0
50	Infosys	41.0	3000.0	Chennai	India	0
57	Infosys	51.0	3184.0	Hyderabad	India	0
68	Infosys	43.0	4034.0	Mumbai	India	0
75	Infosys	44.0	3000.0	Cochin	India	0
85	Infosys	41.0	3000.0	Delhi	India	0
92	Infosys	54.0	3184.0	Mumbai	India	0
103	Infosys	44.0	4034.0	Delhi	India	0
121	Infosys	44.0	3234.0	Mumbai	India	0
128	Infosys	50.0	3184.0	Calcutta	India	0
135	Cognizant	44.0	3033.0	Cochin	India	0
137	Infosys	44.0	4034.0	Hyderabad	India	0
141	Infosys	44.0	4034.0	Delhi	India	1

In [112…

```
#Plot the chart with age and salary
#correlation between salary and age
```

In [113…

```
plt.scatter(df['Age'],df['Salary'],color='g')
plt.xlabel("Age")
plt.ylabel("Salary")
plt.title("Age v/s Salary")
plt.show()
```



Observations from scatter plot:

1. There is no visible correlation between the salary and age of an employee.

In [114…
```
#Count the number of people from each place and represent it visually
```

In [115…
```
df['Place'].value_counts()
```

Out[115…
```
Place
Mumbai         48
Calcutta       32
Chennai        14
Delhi          14
Cochin         13
Noida           8
Hyderabad       8
Pondicherry     3
Pune            2
Bhopal          1
Nagpur          1
Name: count, dtype: int64
```

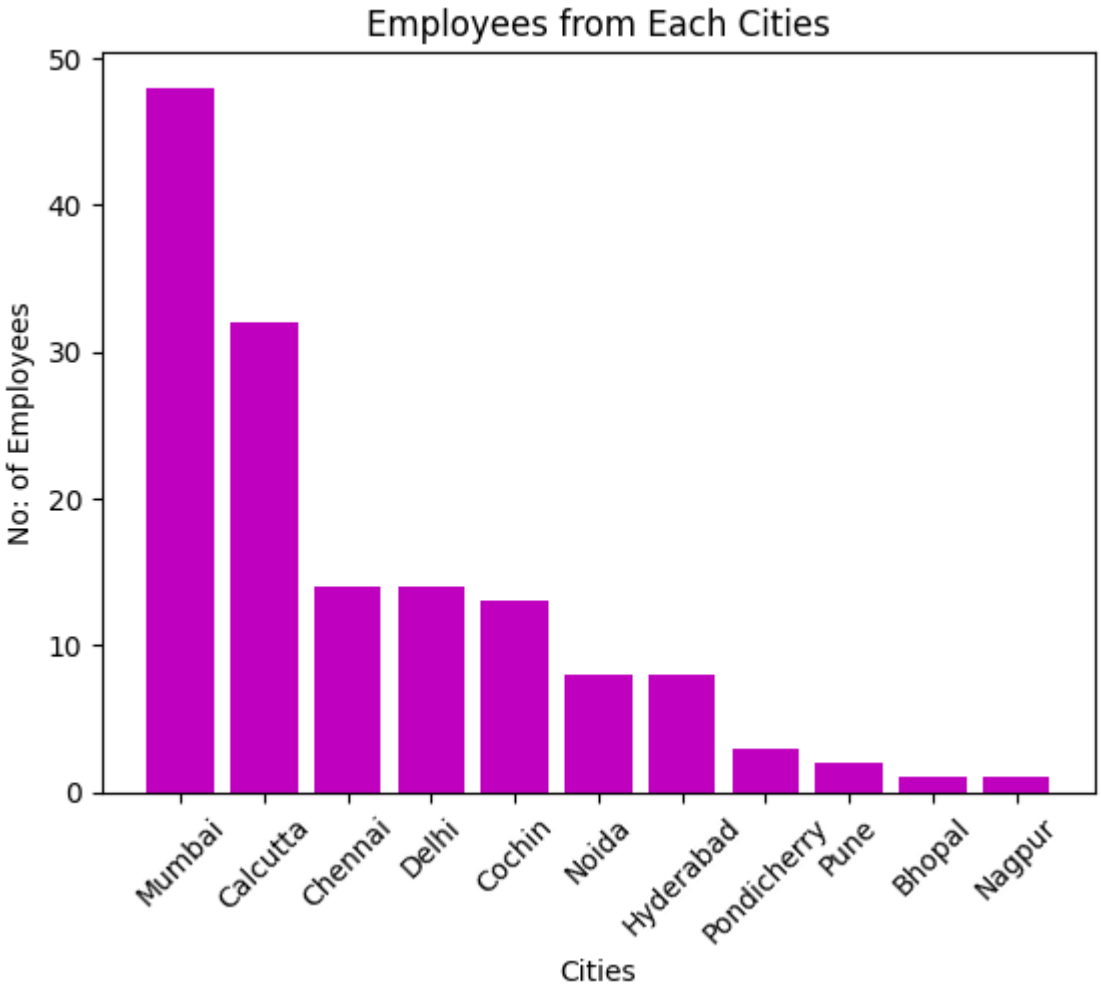
In [116…
```
x=df['Place'].value_counts().index
x
```

Out[116…
```
Index(['Mumbai', 'Calcutta', 'Chennai', 'Delhi', 'Cochin', 'Noida',
       'Hyderabad', 'Pondicherry', 'Pune', 'Bhopal', 'Nagpur'],
      dtype='object', name='Place')
```

In [117…
```
y=df['Place'].value_counts().values
y
```

Out[117…
```
array([48, 32, 14, 14, 13,  8,  8,  3,  2,  1,  1])
```

In [118…
```
plt.bar(x,y,color='m')
plt.xlabel("Cities")
plt.ylabel("No: of Employees")
plt.title("Employees from Each Cities")
plt.xticks(rotation=45)
plt.show()
```



Observations from bar chart:

1. Most of the employees are from Mumbai.
2. Nagpur consisting of the least number of employee

Data Encoding

In [119…
```
#categorical data encoding
```

In [120…
```
df.head(1)
```

Out[120…

	Company	Age	Salary	Place	Country	Gender
0	TCS	20.0	5283.0	Chennai	India	0

In [121…
```
# the 'Gender' column is a 'Nominal categorical' column. so it couldn't be ranke
# Assume M=0 & F=1
```

In [122…

```
df['Gender'].replace(0,'M',inplace=True)
```

```
/tmp/ipykernel_55/1478341521.py:1: FutureWarning: A value is trying to be set on a co
a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because t
intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({
value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the ope
inplace on the original object.


  df['Gender'].replace(0,'M',inplace=True)
```

In [123…

```
df['Gender'].replace(1,'F',inplace=True)
```

In [124…

```
df.head()
```

Out[124…

	Company	Age	Salary	Place	Country	Gender
0	TCS	20.0	5283.0	Chennai	India	M
1	Infosys	30.0	5283.0	Mumbai	India	M
2	TCS	35.0	2300.0	Calcutta	India	M
3	Infosys	40.0	3000.0	Delhi	India	M
4	TCS	23.0	4000.0	Mumbai	India	M

In [125…

```
df['Gender'].value_counts()
```

Out[125…

```
Gender
M    112
F     32
Name: count, dtype: int64
```

In [126…

```
df.describe()
```

Out[126…

	Age	Salary
count	144.000000	144.000000
mean	32.034722	5283.395833
std	8.234681	2368.350171
min	17.000000	1089.000000
25%	23.750000	3045.000000
50%	32.000000	5283.000000
75%	36.000000	7084.000000
max	54.000000	9876.000000

In [127…

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 144 entries, 0 to 143
Data columns (total 6 columns):
 #   Column   Non-Null Count  Dtype
---  ------   --------------  -----
 0   Company  144 non-null    object
 1   Age      144 non-null    float64
 2   Salary   144 non-null    float64
 3   Place    144 non-null    object
 4   Country  144 non-null    object
 5   Gender   144 non-null    object
dtypes: float64(2), object(4)
memory usage: 6.9+ KB
```

In [128…

```
# Here we have 4 categorical columns, "Company", "Place", "Country" & "Gender".
# Four of them are Nominal Categorical columns.
# since only one country's data is available,so we can skip the 'Country' column
```

In [129…

```
df.drop('Country',axis=1,inplace=True)
df
```

Out[129…

	Company	Age	Salary	Place	Gender
0	TCS	20.0	5283.0	Chennai	M
1	Infosys	30.0	5283.0	Mumbai	M
2	TCS	35.0	2300.0	Calcutta	M
3	Infosys	40.0	3000.0	Delhi	M
4	TCS	23.0	4000.0	Mumbai	M
...	...	...	...	...	...
139	Infosys	22.0	8202.0	Mumbai	M
140	TCS	33.0	9024.0	Calcutta	F
141	Infosys	44.0	4034.0	Delhi	F
142	TCS	33.0	5034.0	Mumbai	F
143	Infosys	22.0	8202.0	Cochin	M

144 rows × 5 columns

OneHotEncoding

In [130…

```
#Encoding Categorical datas using OneHotEncoding
```

In [131…

```
df['Company'].unique()
```

Out[131…

```
array(['TCS', 'Infosys', 'Cognizant'], dtype=object)
```

In [132…

```
df['Place'].unique()
```

Out[132…

```
array(['Chennai', 'Mumbai', 'Calcutta', 'Delhi', 'Pondicherry', 'Cochin',
       'Noida', 'Hyderabad', 'Bhopal', 'Nagpur', 'Pune'], dtype=object)
```

In [133…

```
df['Gender'].unique()
```

Out[133…

```
array(['M', 'F'], dtype=object)
```

In [134…

```
#use 'OneHotEncoder' to create dummy variables for each of the nominal categoric
```

In [135…

```
# import libraries
from sklearn.preprocessing import OneHotEncoder
```

In [136…

```
ohe=OneHotEncoder()
```

In [137…

```
ohe
#created an object in the class()
```

Out[137…

```
▾ OneHotEncoder
OneHotEncoder()
```

In [138…

```
#dummies into an array
df_array=ohe.fit_transform(df[['Company','Place','Gender']]).toarray()
df_array
```

Out[138…

```
array([[0., 0., 1., ..., 0., 0., 1.],
       [0., 1., 0., ..., 0., 0., 1.],
       [0., 0., 1., ..., 0., 0., 1.],
       ...,
       [0., 1., 0., ..., 0., 1., 0.],
       [0., 0., 1., ..., 0., 1., 0.],
       [0., 1., 0., ..., 0., 0., 1.]])
```

In [139…

```
ohe.categories_
```

Out[139…

```
[array(['Cognizant', 'Infosys', 'TCS'], dtype=object),
 array(['Bhopal', 'Calcutta', 'Chennai', 'Cochin', 'Delhi', 'Hyderabad',
        'Mumbai', 'Nagpur', 'Noida', 'Pondicherry', 'Pune'], dtype=object),
 array(['F', 'M'], dtype=object)]
```

In [140…

```
categories=[np.array(['Cognizant','Infosys','TCS']),
            np.array(['Bhopal','Calcutta','Chennai','Cochin','Delhi','Hyderabad'
            np.array(['F','M'])]


single_list=[innerlistone for main_array in categories for innerlistone in main_
single_list
#now it's in one list.
```

Out[140…

```
[np.str_('Cognizant'),
 np.str_('Infosys'),
 np.str_('TCS'),
 np.str_('Bhopal'),
 np.str_('Calcutta'),
 np.str_('Chennai'),
 np.str_('Cochin'),
 np.str_('Delhi'),
 np.str_('Hyderabad'),
 np.str_('Mumbai'),
 np.str_('Nagpur'),
 np.str_('Noida'),
 np.str_('Pondicherry'),
 np.str_('Pune'),
 np.str_('F'),
 np.str_('M')]
```

In [141…

```
# create a dataset using the dummies array,'df_array' created using onehotencode
# use 'single_list' for column labelling.
df_new=pd.DataFrame(df_array,columns=single_list)
df_new
```

Out[141…

	Cognizant	Infosys	TCS	Bhopal	Calcutta	Chennai	Cochin	Delhi	Hyderabad	Mumbai
0	0.0	0.0	1.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0
1	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0
2	0.0	0.0	1.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0
3	0.0	1.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0
4	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0
…	…	…	…	…	…	…	…	…	…	…
139	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0
140	0.0	0.0	1.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0
141	0.0	1.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0
142	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0
143	0.0	1.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0

144 rows × 16 columns

In [142…

```
#concat our dataset 'df' with 'df_new'.
```

In [143…

```
df_ml=pd.concat([df,df_new],axis=1)
df_ml
```

Out[143…

	Company	Age	Salary	Place	Gender	Cognizant	Infosys	TCS	Bhopal	Calcutta	…
0	TCS	20.0	5283.0	Chennai	M	0.0	0.0	1.0	0.0	0.0	…
1	Infosys	30.0	5283.0	Mumbai	M	0.0	1.0	0.0	0.0	0.0	…
2	TCS	35.0	2300.0	Calcutta	M	0.0	0.0	1.0	0.0	1.0	…
3	Infosys	40.0	3000.0	Delhi	M	0.0	1.0	0.0	0.0	0.0	…
4	TCS	23.0	4000.0	Mumbai	M	0.0	0.0	1.0	0.0	0.0	…
…	…	…	…	…	…	…	…	…	…	…	…
139	Infosys	22.0	8202.0	Mumbai	M	0.0	1.0	0.0	0.0	0.0	…
140	TCS	33.0	9024.0	Calcutta	F	0.0	0.0	1.0	0.0	1.0	…
141	Infosys	44.0	4034.0	Delhi	F	0.0	1.0	0.0	0.0	0.0	…
142	TCS	33.0	5034.0	Mumbai	F	0.0	0.0	1.0	0.0	0.0	…
143	Infosys	22.0	8202.0	Cochin	M	0.0	1.0	0.0	0.0	0.0	…

144 rows × 21 columns

In [144…

```
#delete the categorical column from dataset 'df_ml'
#df_ml.drop(['index','Company','Place','Gender'],axis=1,inplace=True)
```

In [145…

```
# Final encoded dataset for Machine Learning,
df_ml
```

Out[145…

	Company	Age	Salary	Place	Gender	Cognizant	Infosys	TCS	Bhopal	Calcutta	…
0	TCS	20.0	5283.0	Chennai	M	0.0	0.0	1.0	0.0	0.0	…
1	Infosys	30.0	5283.0	Mumbai	M	0.0	1.0	0.0	0.0	0.0	…
2	TCS	35.0	2300.0	Calcutta	M	0.0	0.0	1.0	0.0	1.0	…
3	Infosys	40.0	3000.0	Delhi	M	0.0	1.0	0.0	0.0	0.0	…
4	TCS	23.0	4000.0	Mumbai	M	0.0	0.0	1.0	0.0	0.0	…
…	…	…	…	…	…	…	…	…	…	…	…
139	Infosys	22.0	8202.0	Mumbai	M	0.0	1.0	0.0	0.0	0.0	…
140	TCS	33.0	9024.0	Calcutta	F	0.0	0.0	1.0	0.0	1.0	…
141	Infosys	44.0	4034.0	Delhi	F	0.0	1.0	0.0	0.0	0.0	…
142	TCS	33.0	5034.0	Mumbai	F	0.0	0.0	1.0	0.0	0.0	…
143	Infosys	22.0	8202.0	Cochin	M	0.0	1.0	0.0	0.0	0.0	…

144 rows × 21 columns

In [ ]:

LabelEncoder

In [146…

```
# example for encoding using LabelEncoder
```

In [147…

```
#save the cleaned dataset 'df' in external
df.to_csv('Employee_cleaned.csv',index=False)
```

In [148…

```
#import library
from sklearn.preprocessing import LabelEncoder
```

In [149…

```
label=LabelEncoder()
print(label)
```

```
LabelEncoder()
```

In [150…

```
df_sample=pd.read_csv('Employee_cleaned.csv')
df_sample['Company']=label.fit_transform(df_sample['Company'])
```

In [151…

```
df_sample
```

Out[151…

	Company	Age	Salary	Place	Gender
0	2	20.0	5283.0	Chennai	M
1	1	30.0	5283.0	Mumbai	M
2	2	35.0	2300.0	Calcutta	M
3	1	40.0	3000.0	Delhi	M
4	2	23.0	4000.0	Mumbai	M
...	...	...	...	...	...
139	1	22.0	8202.0	Mumbai	M
140	2	33.0	9024.0	Calcutta	F
141	1	44.0	4034.0	Delhi	F
142	2	33.0	5034.0	Mumbai	F
143	1	22.0	8202.0	Cochin	M

144 rows × 5 columns

Feature Scaling

In [152…

```
#all numerical columns in the dataset to be standardized before ML model generat
# here we have 'Age' and 'Salary' columns only.
```

In [153…

```
#cleaned dataset
df
```

Out[153…

	Company	Age	Salary	Place	Gender
0	TCS	20.0	5283.0	Chennai	M
1	Infosys	30.0	5283.0	Mumbai	M
2	TCS	35.0	2300.0	Calcutta	M
3	Infosys	40.0	3000.0	Delhi	M
4	TCS	23.0	4000.0	Mumbai	M
...	...	...	...	...	...
139	Infosys	22.0	8202.0	Mumbai	M
140	TCS	33.0	9024.0	Calcutta	F
141	Infosys	44.0	4034.0	Delhi	F
142	TCS	33.0	5034.0	Mumbai	F
143	Infosys	22.0	8202.0	Cochin	M

144 rows × 5 columns

StandardScaler

In [154…

```
# using standard scaler
```

In [155…

```
data=df.iloc[ : ,[1,2]]
data
```

Out[155…

	Age	Salary
0	20.0	5283.0
1	30.0	5283.0
2	35.0	2300.0
3	40.0	3000.0
4	23.0	4000.0
...	...	...
139	22.0	8202.0
140	33.0	9024.0
141	44.0	4034.0
142	33.0	5034.0
143	22.0	8202.0

144 rows × 2 columns

In [156…

```
#import libraries
from sklearn.preprocessing import StandardScaler
```

In [157…

```
scaler=StandardScaler().fit(data)
print(scaler)
```

```
StandardScaler()
```

In [158…

```
#standardize the data
data_scaled=scaler.transform(data)
```

In [159…

```
print(data_scaled)
```

```
[[-1.46656902e+00 -1.67718000e-04]
 [-2.47954254e-01 -1.67718000e-04]
 [ 3.61353128e-01 -1.26409057e+00]
 [ 9.70660511e-01 -9.67494526e-01]
 [-1.10098459e+00 -5.43785893e-01]
 [-4.23130127e-03 -1.20077261e-01]
 [-4.23130127e-03  3.03631371e-01]
 [-1.10098459e+00  7.27340004e-01]
 [ 2.39491652e-01  1.15104864e+00]
 [ 1.57996789e+00  1.57475727e+00]
 [-1.10098459e+00 -1.67718000e-04]
 [ 2.39491652e-01 -1.77720172e+00]
 [ 1.57996789e+00 -1.67718000e-04]
 [-1.71029197e+00 -1.71576397e+00]
 [ 9.70660511e-01 -9.67494526e-01]
 [-1.10098459e+00 -9.67494526e-01]
 [-1.10098459e+00 -9.54783267e-01]
 [ 2.39491652e-01 -1.20077261e-01]
 [-1.22284607e+00 -1.67718000e-04]
 [-4.23130127e-03 -1.67718000e-04]
 [ 6.05076081e-01 -9.48427637e-01]
 [ 2.18927528e+00 -8.89532137e-01]
 [-1.34470754e+00 -1.94649980e-01]
 [-4.23130127e-03  2.33719447e-01]
 [-4.23130127e-03  7.62931529e-01]
 [-1.10098459e+00  1.55060588e+00]
 [ 2.39491652e-01  1.29722811e+00]
 [ 1.57996789e+00  1.69509052e+00]
 [-1.10098459e+00  1.94592603e+00]
 [ 3.61353128e-01 -1.37679706e+00]
 [ 1.70182937e+00  1.00444545e+00]
 [-1.46656902e+00 -9.95459295e-01]
 [ 1.57996789e+00 -5.29379800e-01]
 [ 4.83214605e-01 -1.05671167e-01]
 [-7.35400161e-01  1.23663778e+00]
 [ 3.61353128e-01  1.58492628e+00]
 [-4.23130127e-03 -1.67718000e-04]
 [ 3.61353128e-01 -1.67718000e-04]
 [ 2.39491652e-01 -1.26409057e+00]
 [ 1.09252199e+00 -9.67494526e-01]
 [-9.79123114e-01 -3.97606415e-01]
 [-4.23130127e-03 -1.20077261e-01]
 [-4.23130127e-03  3.03631371e-01]
 [-8.57261637e-01  7.27340004e-01]
 [ 3.61353128e-01  1.15104864e+00]
 [ 1.70182937e+00  1.57475727e+00]
 [-9.79123114e-01 -1.67718000e-04]
 [-4.23130127e-03 -1.77720172e+00]
 [ 1.33624494e+00 -1.67718000e-04]
 [-1.58843050e+00 -1.71576397e+00]
 [ 1.09252199e+00 -9.67494526e-01]
 [-9.79123114e-01 -9.67494526e-01]
 [-1.34470754e+00 -9.54783267e-01]
 [ 3.61353128e-01 -1.20077261e-01]
 [-1.34470754e+00  5.34128867e-01]
 [-4.23130127e-03  1.00444545e+00]
 [ 7.26937558e-01 -9.48427637e-01]
 [ 2.31113675e+00 -8.89532137e-01]
 [-1.10098459e+00 -1.94649980e-01]
 [-4.23130127e-03  2.33719447e-01]
```

```
[-4.23130127e-03  7.62931529e-01]
[-8.57261637e-01  1.55060588e+00]
[ 4.83214605e-01  1.29722811e+00]
[ 1.09252199e+00  1.69509052e+00]
[-8.57261637e-01  5.33705159e-01]
[-1.26092778e-01 -1.37679706e+00]
[ 1.09252199e+00 -1.67718000e-04]
[-1.34470754e+00 -9.95459295e-01]
[ 1.33624494e+00 -5.29379800e-01]
[-4.23130127e-03 -1.05671167e-01]
[-1.34470754e+00  1.23663778e+00]
[ 2.39491652e-01  1.58492628e+00]
[-9.79123114e-01 -1.67718000e-04]
[ 2.39491652e-01 -1.67718000e-04]
[ 6.05076081e-01 -1.26409057e+00]
[ 1.45810642e+00 -9.67494526e-01]
[-4.23130127e-03 -5.43785893e-01]
[-4.23130127e-03 -1.20077261e-01]
[-4.23130127e-03  3.03631371e-01]
[-7.35400161e-01  7.27340004e-01]
[-4.23130127e-03  1.15104864e+00]
[ 1.33624494e+00  1.57475727e+00]
[-1.22284607e+00 -1.67718000e-04]
[ 1.17630175e-01 -1.77720172e+00]
[-1.83215345e+00 -1.71576397e+00]
[ 1.09252199e+00 -9.67494526e-01]
[-1.34470754e+00 -9.67494526e-01]
[-4.23130127e-03 -9.54783267e-01]
[ 4.83214605e-01 -1.20077261e-01]
[-1.34470754e+00 -1.67718000e-04]
[ 2.39491652e-01 -1.67718000e-04]
[-4.23130127e-03 -9.48427637e-01]
[ 2.67672118e+00 -8.89532137e-01]
[-1.22284607e+00 -1.94649980e-01]
[-4.23130127e-03  2.33719447e-01]
[-4.23130127e-03  7.62931529e-01]
[-1.22284607e+00  1.55060588e+00]
[ 1.17630175e-01  1.29722811e+00]
[ 1.45810642e+00  1.69509052e+00]
[-1.22284607e+00 -1.67718000e-04]
[-4.23130127e-03 -1.37679706e+00]
[ 1.45810642e+00 -1.67718000e-04]
[-1.22284607e+00 -9.95459295e-01]
[ 1.45810642e+00 -5.29379800e-01]
[ 1.17630175e-01 -1.05671167e-01]
[-1.22284607e+00  1.23663778e+00]
[-4.23130127e-03  1.58492628e+00]
[ 1.17630175e-01 -1.67718000e-04]
[ 1.17630175e-01 -1.67718000e-04]
[ 1.17630175e-01 -1.26409057e+00]
[-4.23130127e-03 -8.68346706e-01]
[-1.22284607e+00 -4.06504296e-01]
[-4.23130127e-03  6.42359941e-02]
[-4.23130127e-03  1.15081030e-01]
[-1.22284607e+00  1.48450733e+00]
[ 1.17630175e-01  1.48450733e+00]
[ 1.45810642e+00  1.94592603e+00]
[-1.22284607e+00 -1.67718000e-04]
[ 1.17630175e-01 -1.77720172e+00]
[ 1.45810642e+00 -1.67718000e-04]
```

```
 [-4.23130127e-03 -1.71576397e+00]
 [ 1.45810642e+00 -8.68346706e-01]
 [-4.23130127e-03 -7.75130807e-01]
 [-1.22284607e+00  1.48450733e+00]
 [ 1.17630175e-01  1.57028185e-01]
 [-1.46656902e+00 -1.16263883e-01]
 [-2.47954254e-01 -7.85538150e-02]
 [ 1.17630175e-01 -9.48427637e-01]
 [ 2.18927528e+00 -8.89532137e-01]
 [-4.23130127e-03  7.62931529e-01]
 [-1.22284607e+00  1.55060588e+00]
 [-1.26092778e-01  1.29722811e+00]
 [ 9.70660511e-01  1.69509052e+00]
 [-4.23130127e-03 -1.67718000e-04]
 [-1.26092778e-01 -1.37679706e+00]
 [ 1.45810642e+00 -9.53512141e-01]
 [-1.22284607e+00 -9.95459295e-01]
 [ 1.45810642e+00 -5.29379800e-01]
 [ 1.17630175e-01 -1.05671167e-01]
 [-1.22284607e+00  1.23663778e+00]
 [ 1.17630175e-01  1.58492628e+00]
 [ 1.45810642e+00 -5.29379800e-01]
 [ 1.17630175e-01 -1.05671167e-01]
 [-1.22284607e+00  1.23663778e+00]]
```

In [160…

```python
#check the mean, for a standarzide data it's = 0 or near,
print(data_scaled.mean(axis=0))
```

```
[8.17247504e-17 1.03312420e-16]
```

In [161…

```python
#check the std, for a standarzide data it's = 1,
print(data_scaled.std(axis=0))
```

```
[1. 1.]
```

In [162…

```python
#convert it into dataframe-table format
scaled_data_set=pd.DataFrame(data_scaled,columns=data.columns)
scaled_data_set
```

Out[162…

	Age	Salary
0	-1.466569	-0.000168
1	-0.247954	-0.000168
2	0.361353	-1.264091
3	0.970661	-0.967495
4	-1.100985	-0.543786
…	…	…
139	-1.222846	1.236638
140	0.117630	1.584926
141	1.458106	-0.529380
142	0.117630	-0.105671
143	-1.222846	1.236638

144 rows × 2 columns

In [ ]:

MinMaxScaler

In [163…
```
# using minmax scaler
```

In [164…
```
data=df.iloc[ : ,[1,2]]
data
```

Out[164…

	Age	Salary
0	20.0	5283.0
1	30.0	5283.0
2	35.0	2300.0
3	40.0	3000.0
4	23.0	4000.0
…	…	…
139	22.0	8202.0
140	33.0	9024.0
141	44.0	4034.0
142	33.0	5034.0
143	22.0	8202.0

144 rows × 2 columns

In [165…
```
#import libraries
from sklearn.preprocessing import MinMaxScaler
```

In [166…
```
scaler=MinMaxScaler().fit(data)
print(scaler)
```

```
MinMaxScaler()
```

In [167…
```
#check min and max
scaler.data_min_
```

Out[167…
```
array([  17., 1089.])
```

In [168…
```
scaler.data_max_
```

Out[168…
```
array([  54., 9876.])
```

In [169…
```
data.describe()
```

Out[169…

	Age	Salary
count	144.000000	144.000000
mean	32.034722	5283.395833
std	8.234681	2368.350171
min	17.000000	1089.000000
25%	23.750000	3045.000000
50%	32.000000	5283.000000
75%	36.000000	7084.000000
max	54.000000	9876.000000

In [170…

```
#check feature range
scaler.feature_range
```

Out[170…

```
(0, 1)
```

In [171…

```
#standardize the data
data_scaled_mm=scaler.transform(data)
```

In [172…

```
print(data_scaled_mm)
```

```
[[0.08108108 0.47729601]
 [0.35135135 0.47729601]
 [0.48648649 0.13781723]
 [0.62162162 0.21748037]
 [0.16216216 0.33128485]
 [0.40540541 0.44508934]
 [0.40540541 0.55889382]
 [0.16216216 0.6726983 ]
 [0.45945946 0.78650279]
 [0.75675676 0.90030727]
 [0.16216216 0.47729601]
 [0.45945946 0.        ]
 [0.75675676 0.47729601]
 [0.02702703 0.01650165]
 [0.62162162 0.21748037]
 [0.16216216 0.21748037]
 [0.16216216 0.2208945 ]
 [0.45945946 0.44508934]
 [0.13513514 0.47729601]
 [0.40540541 0.47729601]
 [0.54054054 0.22260157]
 [0.89189189 0.23842039]
 [0.10810811 0.42505975]
 [0.40540541 0.54011608]
 [0.40540541 0.68225788]
 [0.16216216 0.89382042]
 [0.45945946 0.82576534]
 [0.75675676 0.93262775]
 [0.16216216 1.        ]
 [0.48648649 0.10754524]
 [0.78378378 0.74712644]
 [0.08108108 0.20996927]
 [0.75675676 0.33515421]
 [0.51351351 0.44895869]
 [0.24324324 0.80949129]
 [0.48648649 0.90303858]
 [0.40540541 0.47729601]
 [0.48648649 0.47729601]
 [0.45945946 0.13781723]
 [0.64864865 0.21748037]
 [0.18918919 0.3705474 ]
 [0.40540541 0.44508934]
 [0.40540541 0.55889382]
 [0.21621622 0.6726983 ]
 [0.48648649 0.78650279]
 [0.78378378 0.90030727]
 [0.18918919 0.47729601]
 [0.40540541 0.        ]
 [0.7027027  0.47729601]
 [0.05405405 0.01650165]
 [0.64864865 0.21748037]
 [0.18918919 0.21748037]
 [0.10810811 0.2208945 ]
 [0.48648649 0.44508934]
 [0.10810811 0.62080346]
 [0.40540541 0.74712644]
 [0.56756757 0.22260157]
 [0.91891892 0.23842039]
 [0.16216216 0.42505975]
 [0.40540541 0.54011608]
```

```
[0.40540541 0.68225788]
[0.21621622 0.89382042]
[0.51351351 0.82576534]
[0.64864865 0.93262775]
[0.21621622 0.62068966]
[0.37837838 0.10754524]
[0.64864865 0.47729601]
[0.10810811 0.20996927]
[0.7027027  0.33515421]
[0.40540541 0.44895869]
[0.10810811 0.80949129]
[0.45945946 0.90303858]
[0.18918919 0.47729601]
[0.45945946 0.47729601]
[0.54054054 0.13781723]
[0.72972973 0.21748037]
[0.40540541 0.33128485]
[0.40540541 0.44508934]
[0.40540541 0.55889382]
[0.24324324 0.6726983 ]
[0.40540541 0.78650279]
[0.7027027  0.90030727]
[0.13513514 0.47729601]
[0.43243243 0.        ]
[0.         0.01650165]
[0.64864865 0.21748037]
[0.10810811 0.21748037]
[0.40540541 0.2208945 ]
[0.51351351 0.44508934]
[0.10810811 0.47729601]
[0.45945946 0.47729601]
[0.40540541 0.22260157]
[1.         0.23842039]
[0.13513514 0.42505975]
[0.40540541 0.54011608]
[0.40540541 0.68225788]
[0.13513514 0.89382042]
[0.43243243 0.82576534]
[0.72972973 0.93262775]
[0.13513514 0.47729601]
[0.40540541 0.10754524]
[0.72972973 0.47729601]
[0.13513514 0.20996927]
[0.72972973 0.33515421]
[0.43243243 0.44895869]
[0.13513514 0.80949129]
[0.40540541 0.90303858]
[0.43243243 0.47729601]
[0.43243243 0.47729601]
[0.43243243 0.13781723]
[0.40540541 0.24411062]
[0.13513514 0.36815751]
[0.40540541 0.49459429]
[0.40540541 0.50825083]
[0.13513514 0.87606692]
[0.43243243 0.87606692]
[0.72972973 1.        ]
[0.13513514 0.47729601]
[0.43243243 0.        ]
[0.72972973 0.47729601]
```

```
[0.40540541 0.01650165]
[0.72972973 0.24411062]
[0.40540541 0.2691476 ]
[0.13513514 0.87606692]
[0.43243243 0.51951747]
[0.08108108 0.44611358]
[0.35135135 0.45624218]
[0.43243243 0.22260157]
[0.89189189 0.23842039]
[0.40540541 0.68225788]
[0.13513514 0.89382042]
[0.37837838 0.82576534]
[0.62162162 0.93262775]
[0.40540541 0.47729601]
[0.37837838 0.10754524]
[0.72972973 0.22123592]
[0.13513514 0.20996927]
[0.72972973 0.33515421]
[0.43243243 0.44895869]
[0.13513514 0.80949129]
[0.43243243 0.90303858]
[0.72972973 0.33515421]
[0.43243243 0.44895869]
[0.13513514 0.80949129]]
```

In [173…

```
#convert it into dataframe-table format
mm_scaled_data_set=pd.DataFrame(data_scaled_mm,columns=data.columns)
mm_scaled_data_set
```

Out[173…

	Age	Salary
0	0.081081	0.477296
1	0.351351	0.477296
2	0.486486	0.137817
3	0.621622	0.217480
4	0.162162	0.331285
...	...	...
139	0.135135	0.809491
140	0.432432	0.903039
141	0.729730	0.335154
142	0.432432	0.448959
143	0.135135	0.809491

144 rows × 2 columns

In [174…

```
# The dataset is now completely preprocessed for Machine Learning Modeling
```